

Professional Summary

Device Architect & Embedded Firmware Lead with 5.6+ years of experience in medical devices, industrial automation, and advanced control systems, driving end-to-end development of safety-critical, standards-compliant products from concept to production.

Core Competencies

- **Embedded Systems & Firmware**
 - Device Architecture & End-to-End Product Development
 - Hardware Design, Schematics and Board Bringup
 - Embedded C/C++ (Bare-Metal), Interrupt-Driven Design
 - Bootloader, OTA & Runtime Self-Programming
 - Flash Memory & SDHC Storage Systems
 - Microcontrollers: PIC18/24/32, STM32, ESP32, Portenta H7
 - Communication Protocols: UART, I2C, SPI, QSPI, CAN
 - Timing Sequence Execution & System Validation
 - Network Packet Analysis, Wireshark, Wi-Fi Debugging (Sniffer)
 - IoT Devices Management
- **Medical & Safety-Critical Systems**
 - Respiratory Therapy Systems
 - ISO 80601-2-70
 - ISO 80601-2-72
 - IEC 62304
 - IEC 60601-1
 - IEC 60601-2
 - Safety-Critical Firmware & Validation
- **Control Systems & Algorithms**
 - Closed-Loop Control (PID, Feedforward, Fuzzy Logic)
 - Pressure Control & Therapy Algorithm Development
 - BLDC, Servo, Stepper & DC Motor Control
 - Motion Control Algorithms & Mechanical Calculations
 - Sensor Calibration, Filtering & Measurement Techniques
 - IMU Integration & Kalman Filter
 - Machine Learning(Prediction and Prevention Algorithms)
 - Computer Vision
- **Tools & Software**
 - MPLAB X
 - GX Works2/3, GT Designer 3
 - MR Configurator2
 - SolidWorks
 - VS Code, PyCharm
 - Git & SourceTree
- **Industrial Automation & Mechatronics**
 - PLC Programming (IEC 61131) & HMI Design
 - Industrial Automation & Relay Logic
 - Pneumatics & Hydraulics (Festo Systems)
 - Electrical Circuit Design & BOM Management
 - Servo Drives, VFDs & Proportional Valves
 - 3D Printing & SolidWorks Concept Design
- **Professional Strengths**
 - Cross-Functional Coordination (UI/UX, Software, Sales, Production, PCB, Hardware, QA)
 - System-Level Troubleshooting
 - Vendor Management & Cost Optimization
 - Multidisciplinary Mechatronics Integration

Professional Experience - 5.6+ Years

- **Embedded Engineer(Firmware Team Lead), Avantel (August 2023 – Present) , Vishakaptnam & Hyderabad**
 - Device Architect & Firmware/Firmware Team Lead for CPAP, BiPAP, and HFNC medical devices, driving end-to-end product development from concept to production while ensuring compliance with ISO 80601-2-70 and IEC 62304 standards.
 - Designed and implemented therapy control and safety algorithms, including CPAP, APAP, BIPAP (S/T/ST) and HFNC, improving patient comfort and device performance.
 - Developed real-time embedded firmware (bare-metal C) on PIC18F/24F/32, including advanced blower motor control using closed-loop, feedforward, and fuzzy logic.
 - Architected device–app–cloud connectivity, enabling Wi-Fi data transfer, therapy logging, and OTA firmware updates with fail-safe mechanisms.

- Led sensor calibration, hardware–firmware integration, GUI development, and advanced system-level debugging for clinical-grade accuracy.
- Designed in-house test and simulation devices (snoring & breath simulators), saving the company 12+ lakhs and reducing dependency on external equipment.
- Implemented Algorithms to detect Snoring and breathing and flow limitation patterns.

➤ **Senior Engineer, HBL Power Systems Limited (September 2021 – April 2023), Hyderabad**

- Led cross-functional collaboration to design and deploy in-house industrial automation systems, reducing dependency on external vendors and improving system turnaround time.
- Drove Industry 4.0 initiatives, implementing automation and process optimization strategies to enhance operational efficiency.
- Designed electrical and electronic systems including BOM preparation, circuit development, and actuator–sensor integration for machine development.
- Developed control logic and firmware for microcontrollers, PLCs (IEC 61131), and HMIs, ensuring reliable machine performance and validation.
- Delivered multiple cost-effective automation solutions by optimizing hardware selection and system architecture.
- Designed and deployed a cloud-based RH monitoring solution for glove boxes, achieving ~90% cost reduction compared to external vendor quotation.

➤ **Graduate Engineer Trainee-Contract, VSRMKP Technologies Private Limited (January 2021 – June 2021)**

- Designed a Li-ion battery pack and built an E-bike prototype using a BLDC hub motor.
- Developed embedded software for monitoring battery parameters like voltage, current, and temperature.
- Designed BMS circuits with over-voltage, under-voltage, and reverse polarity protection.
- Assembled and tested battery pack with E-Bike.

➤ **Technical Assistant-Contract, Mahatma Gandhi Institute of Technology (October 2019 – April 2020)**

- Designed and fabricated an Automatic TIG Welding system using mechatronics concepts.
- Integrated DC gear motor, sensors, encoders, and microcontroller code for constant motion in welding operation.
- This project was sponsored by JNTUH (JNTUH/TEQIP-III/CRS/2019/MECH/06).

Projects

1. CPAP/BiPAP/AVAPS
2. Breath/Snoring Simulators
3. Computer Vision - Colour Sorting / Neck hole detection with mm calculations
4. IoT - Device data push to cloud, Device Clustering.
5. Li-Ion Battery Management System (1S–7S BMS) for E-Bike Application
6. BLDC Hub Motor E-Bike Prototype Development & System Integration
7. PLC & HMI Based Projects(Tank Level, Weight sorting)
8. Low Cost Automation(Relay Level)
9. Control System Algorithms(Pressure, Flow, Volume control).
10. Model Training/Prediction(IMU) - Guster prediction.

Education

B.Tech – Mechatronics

Mahatma Gandhi Institute of Technology – JNTU Hyderabad – 2019

Master Certificate – Mechatronics (NSQF Level 6)

Central Institute of Tool Design – 2020

Awards

- 1st Prize, State Level – INSPIRE for Robo Boat
- Presented project **Advanced Robot**, NIT Warangal
- Secured 1st Merit in MGIT MAGISTECH Robotic (All Terrain)
- Poster Presentation in 69th AGM and National Conference on Unmanned Aircraft Systems- “PATH PLANNING AND NAVIGATION”.
- Received recognition in the "Innovative Student Project Award" for the Path-Correcting Multipurpose Agriculture Robot, which was selected for presentation at the national level by INSA and INSE

Hyperlink-attached

- [Detaled work info](#)
- [Projects](#)
- [CV](#)

Declaration

I hereby declare that the details and information given above are complete and true to the best of my knowledge.

Place: Hyderabad

Date: 25-May-2026

(Vamshi Oruganti)